

Ex2 Task 4.

σ^2 is the part correspond to aleatoric because it is the original data noises and therefore does not change because it is part of the data.

$\phi_A^T \Sigma_{\text{post}} \phi_A$ is the epistemic uncertainty part. This is because it is the posterior variance which is the uncertainty of weight.

The former does not shrink when $n \rightarrow \infty$ because it is part of data and cannot be changed. The latter change because as $n \rightarrow \infty$, uncertainty decrease, thus it shrinks.